ALGEBRA 2 · SEQUENCES & SERIES

Sequence Builder

Explore & Reflect

Name _____ Date _____ Period _____

Open **Sequence Builder** on Mr. K's Classroom: Algebra 2 → Sequences & Series → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Starting at 1, which grows faster: adding 5 each time or multiplying by 2 each time?

_____

_____

2 Explore use the tool and record what you see

Identify each sequence

Rebuild each sequence in the tool.

Sequence	Type	d or r	Next term	Formula
3, 7, 11, 15, …				
5, 10, 20, 40, …				
20, 17, 14, 11, …				
8, −4, 2, −1, …				
16, 8, 4, 2, …				

Running sums for $a_1 = 8$, $r = \frac{1}{2}$

n	1	2	3	4	5	→ ∞
S_n	8					

3 Write the rule fill in each blank from what you observed

1. Arithmetic sequences _____ the same number each time; their terms lie on a _____.
2. Geometric sequences _____ by the same number each time; their terms follow an _____ curve.
3. A negative common ratio makes the terms _____ signs.
4. An infinite geometric series has a sum only when _____, and then the sum is _____.

4 Check no tool: show your work

1. Find the 10th term of 4, 9, 14, …

2. Find the 6th term of 3, 6, 12, …

3. Find the sum of 10 + 5 + 2.5 + …

4. Does 2 + 6 + 18 + … have a sum?

5 Reflect

Why does a geometric series with $r = \frac{1}{2}$ stop growing past a certain value?